Yunke Wang

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Qualifications

The University of Sydney (USYD) — Sydney, Australia
Postdoctoral Research Associate; Advisor: Prof. Chang Xu — *Aug. 2024 – Present*

Wuhan University (WHU) — Wuhan, China
Ph.D. in Artificial Intelligence; Advisor: Prof. Bo Du — *Sept. 2019 – June 2024*

Wuhan University (WHU) — Wuhan, China
B.Eng. in Computer Science — *Sept. 2015 – June 2019*

Research Interests

Embodied AI, Generative Models, Computer Vision, Machine Learning

Publications

First-author papers

[9] **Yunke Wang**, Linwei Tao, Yutian Lin, Chang Xu, Bo Du. "Visual Adversarial Imitation Learning with Calibrated Contrastive Representation,"
Journal of Image and Graphics (CJIG), 2026. **CCF B**

[8] **Yunke Wang**, Chang Xu, Tianyu Guo, Bo Du, Dacheng Tao. "On Positive-Unlabeled Classification from Corrupted Data in GANs,"
IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2025. **CCF A**

[7] **Yunke Wang**, Yanxi Li, Chang Xu. "Position: AI Scaling: From Up to Down and Out,"
International Conference on Machine Learning (ICML), 2025. **CCF A**

[6] **Yunke Wang**, Minjing Dong, Yukun Zhao, Chang Xu, Bo Du. "Imitation Learning from Purified Demonstrations,"
International Conference on Machine Learning (ICML), 2024. **CCF A**

[5] **Yunke Wang**, Wenyuan Wang, Bo Du, Chang Xu. "Multi-Tailed Vision Transformer for Efficient Inference,"
Neural Networks, 2024. **CCF B**

[4] **Yunke Wang**, Xiyu Wang, Anh-Dung Dinh, Bo Du, Chang Xu. "Learning to Schedule in Diffusion Probabilistic Models,"
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2023. **CCF A**

[3] **Yunke Wang**, Bo Du, Chang Xu. "Unlabeled Imperfect Demonstrations in Adversarial Imitation Learning,"
AAAI Conference on Artificial Intelligence (AAAI), 2023. **CCF A**

[2] **Yunke Wang**, Chang Xu, Bo Du, Honglak Lee. "Learning to Weight Imperfect Demonstrations,"
International Conference on Machine Learning (ICML), 2021. **CCF A**

[1] **Yunke Wang**, Chang Xu, Bo Du. "Robust Adversarial Imitation Learning via Adaptively-Selected Demonstrations,"
International Joint Conference on Artificial Intelligence (IJCAI), 2021. **CCF A**

Other papers

[27] Jingxuan He, Xiyu Wang, **Yunke Wang**, Mengyu Zheng, Chang Xu. "Early Semantic Grounding in Image Editing Models for Zero-Shot Referring Image Segmentation,"
Annual Conference on Neural Information Processing Systems (NeurIPS), 2026. **CCF A**

[26] Yutong Wang, **Yunke Wang**, Tianfan Xue, Yu Qiao, Yaohui Wang, Xinyuan Chen, Chang Xu. " PARE: Pruning and Adaptive Routing for Efficient Video Generation,"
Annual Conference on Neural Information Processing Systems (NeurIPS), 2026. **CCF A**

[25] Laibin Chang, **Yunke Wang**, Longxiang Deng, Jiaxing Huang, Bo Du, Chang Xu. "Marine Saliency Segmenter: Object-Focused Conditional Diffusion with Region-Level Semantic Knowledge Distillation."
IEEE Transactions on Image Processing (TIP), 2026. **CCF A**

[24] Laibin Chang, **Yunke Wang**, Bo Du, Chang Xu. "Color Correction Meets Cross-Spectral Refinement: A Distribution-Aware Diffusion for Underwater Image Restoration,"
IEEE Transactions on Multimedia (TMM), 2026. **CCF A**

[23] Chenghao Xia, Siyu Xu, Chengbin Du, **Yunke Wang**. "RAAG: Retrieval-Augmented Action Generation."
International Conference on Digital Image Computing: Techniques and Applications (DICTA), 2026.

[22] Enhuai Liu, Changmin Sun, **Yunke Wang**. "PASTA: Faster Video Diffusion Transformer with Partial Spatial-Temporal Attention."
International Conference on Digital Image Computing: Techniques and Applications (DICTA), 2026.

[21] Eason Yu, Tzu Hao Liu, Clement L. Canonne, **Yunke Wang**, Chang Xu, Nguyen H. Tran, Stefano V. Albrecht. "NashPG: A Policy Gradient Method with Iteratively Refined Regularization for Finding Nash Equilibria."
Transactions on Machine Learning Research (TMLR), 2026.

[20] Jingxuan He, Xiyu Wang, Mengyu Zheng, Xiangyu Zeng, **Yunke Wang**, Chang Xu. "Rethinking Where to Edit: Task-Aware Localization for Instruction-Based Image Editing,"
ACM International Conference on Multimedia (ACM MM), 2026. **CCF A**

[19] Hongbo Gao, Zhengyu Li, Xueru Nie, Dihao Zhu, Lijun Zhao, **Yunke Wang**, Chang Xu. "RA-SOD: Reliability-Aware RGB-T Salient Object Detection under Modality Degradation,"
European Conference on Computer Vision (ECCV), 2026. **CCF B**

[18] Weiying Xie, Qianchen Zeng, Zihan Meng, Jiajun Tian, Sibo He, Danian Yang, Jie Du, **Yunke Wang**, Daixun Li, Hengyi Wang, Jitao Ma, Leyuan Fang, Yunsong Li. "Towards Efficient Embodied Reasoning: Mixture-of-Depth Compute Allocation for Vision-Language-Action Model,"
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2026. **CCF A**

[17] Chenyu Hui, Xiaodi Huang, Siyu Xu, **Yunke Wang**, Shan You, Tao Huang, Chang Xu. "Seeing Realism from Simulation: Efficient Video Transfer for Vision-Language-Action Data Augmentation,"
International Conference on Machine Learning (ICML), 2026. **CCF A**

[16] Xiangyu Zeng, Qi Xu, **Yunke Wang**, Chang Xu. "HiCI: Hierarchical Construction–Integration for Long-Context Attention,"
International Conference on Machine Learning (ICML), 2026. **CCF A**

[15] Yixu Feng, Zinan Zhao, Yanxiang Ma, Chenghao Xia, Chengbin Du, **Yunke Wang**, Chang Xu. "See What Matters: Differentiable Grid Sample Pruning for Generalizable Vision-Language-Action Model,"
International Conference on Machine Learning (ICML), 2026. **CCF A**

[14] Sibo He, Weiying Xie, Daixun Li, Junhao Zhong, Jiayun Tian, **Yunke Wang**, Leyuan Fang, Gang He, Yunsong Li. "Motion Dynamics Learning for Few-Shot Embodied Adaptation,"
International Conference on Machine Learning (ICML), 2026. **CCF A**

[13] Zixuan Huang, Yanxiang Ma, Luhan Wang, **Yunke Wang**, Duo Shi, Chang Xu. "Mitigating Legal Hallucinations via Symbolic Constraints and Analogical Precedents,"
Annual Meeting of the Association for Computational Linguistics (ACL), 2026. **CCF A**

[12] Daixun Li, Zirui Li, Sibo He, Jiayun Tian, Mingxiang Cao, Weiying Xie, **Yunke Wang**, Yusi Zhang, Yunsong Li, Chang Xu, Leyuan Fang. "GeoCoT: Towards Reliable Remote Sensing Reasoning with Manifold Perspective,"
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026, Highlight. **CCF A**

[11] Siyu Xu, Zijian Wang, **Yunke Wang**, Chenghao Xia, Tao Huang, Chang Xu. "Affordance Field Intervention: Enabling VLAs to Escape Memory Traps in Robotic Manipulation,"
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026. **CCF A**

[10] Enhuai Liu, **Yunke Wang**, Changming Sun, Chang Xu. "D2Cache: Second-Order Delta Caching for Higher Video Diffusion Acceleration,"
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026. **CCF A**

[9] Xiaohuan Pei, Yuxing Chen, Siyu Xu, **Yunke Wang**, Yuheng Shi, Chang Xu. "Action-aware Dynamic Pruning for Efficient Vision-Language-Action Manipulation,"
International Conference on Learning Representations (ICLR), 2026. **CCF A**

[8] Rui Xu, **Yunke Wang**, Yong Luo, Bo Du. "Rethinking Visual Token Reduction in LVLMs under Cross-modal Misalignment,"
AAAI Conference on Artificial Intelligence (AAAI), 2026. **CCF A**

[7] Siyu Xu, **Yunke Wang**, Chenghao Xia, Dihao Zhu, Tao Huang, Chang Xu. "VLA-Cache: Efficient Vision-Language-Action Manipulation via Adaptive Token Caching,"
Annual Conference on Neural Information Processing Systems (NeurIPS), 2025. **CCF A**

[6] Daixun Li, Weiying Xie, Mingxiang Cao, **Yunke Wang**, Yusi Zhang, Leyuan Fang, Yunsong Li, Chang Xu. "FusionSAM: Visual Multi-Modal Learning with Segment Anything Model,"
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2025. **CCF A**

[5] Laibin Chang, **Yunke Wang**, Longxiang Deng, Bo Du, Chang Xu. "WaterDiffusion: Learning a Prior-involved Unrolling Diffusion for Joint Underwater Saliency Detection and Visual Restoration,"
AAAI Conference on Artificial Intelligence (AAAI), 2025. **CCF A**

[4] Siyu Xu, **Yunke Wang**, Daochang Liu, Bo Du, Chang Xu. "CollagePrompt: A Benchmark for Budget-Friendly Visual Recognition with GPT-4V",
Findings of the Association for Computational Linguistics (NAACL), 2025. **CCF B**

[3] Daixun Li, Weiying Xie, Leyuan Fang, **Yunke Wang**, Mingxiang Cao, Jitao Ma, Yunsong Li, Chang Xu. "MMFormer: Multi-Modality Semi-Supervised Vision Transformer in Remote Sensing Imagery Classification,"
Neural Networks, 2025. **CCF B**

[2] Rui Xu, **Yunke Wang**, Lihua Zhang, Yong Luo, Shu Yang, Yihui Wang, Bo Du, Hao Chen. "Masked Diffusion Models for Unsupervised Anomaly Detection in Brain Images,"
IEEE International Conference on Bioinformatics and Biomedicine (BIBM), 2025. **CCF B**

[1] Laibin Chang, **Yunke Wang**, Bo Du, Chang Xu. "Rectangling and Enhancing Underwater Stitched Image via Content-aware Warping and Perception Balancing,"
Neural Networks, 2024. **CCF B**

Honors & Awards

Gold Reviewer Award, ICML — May 2026
EMCRs Training Scheme Award, University of Sydney — March 2026
Academic Innovation Scholarship, Wuhan University — Nov. 2025 / 2024 / 2023
Outstanding Graduates, Wuhan University — June 2024
National Scholarship, Ministry of Education (Top 1%) — Nov. 2023
KDD-23 Student Travel Award, KDD — June 2023
AAAI-23 Student Scholarship, AAAI — Jan. 2023
First Prize Winner of China Graduate AI Innovation Competition (8/1174) — Dec. 2022
Yisheng Ph.D Scholarship, Wuhan University — Dec. 2021

Invited Talks

"Vision-Language-Action Model: Bridging the Intelligence to the Physical World" — Feb. 2026
University of Sydney, 2026 S1 Deep Learning Class Week 1.

"Calibration of AI Models for Reliable Decision-Making" Nov. 2025
Commonwealth Bank of Australia, Redfern Office, USYD-CommBank Workshop.

"Policy Learning in Diffusion Models" Nov. 2024
University of Queensland, Brisbane City Campus, International Workshop on Weakly Supervised Learning.

"Imitation Learning from Imperfect Demonstrations" Jun. 2023
"DiDi" Forum for Doctoral Students.

"Dynamic Neural Network in Vision Transformer" Feb. 2023
University of Sydney, 2023 S1 Deep Learning Class Week 1.

"Deep Reinforcement Learning in Autonomous Driving" Oct. 2021
In-Driving Co., Ltd.

"Imitation Learning from Imperfect Demonstrations" Sep. 2021
Doctoral Innovation Forum on Artificial Intelligence in Central and Western China.

Grants & Funding

Enabling VLA Assistive Robotics for the Future of Aged Care in Australia 2026
ARC Linkage Project, LP25 Round 2
Role: Co-CI (Ranked 3rd); Funding: AUD $598,177.

Last-link Automation in Daily Retail Operations 2026
AEA (Australia's Economic Accelerator) Ignite Round 2
Role: Co-Lead Entrepreneur (Ranked 2nd); Funding: AUD $484,498.

AI Coding for Research Productivity: An EMCR Training Workshop 2026
EMCRs Training Scheme Funding, The University of Sydney
Role: Lead Applicant; Funding: AUD $5,000.

Academic Services & Teaching

Academic Leadership:
- Co-Lead of Green Computing Pillar, Net Zero Institute, The University of Sydney, 2024 - Present

Conference Organizer:
- Web Chair of ACM MM [CCF A/CORE A*], 2027
- Local Arrangement Chair of DICTA [Premier Conference on Computer Vision in Australia], 2026

Workshop Organizer:
- AI Coding for Research Productivity: An EMCR Training Workshop, The University of Sydney, 2026
- Sydney AI Meetup - CVPR · ICML Pre-Conference 2026, The University of Sydney, 2026
- Sydney AI Meetup - Symposium 2025, The University of Sydney, 2025
- Workshop on Sustainable AI for the Future Web, The Web Conference [CCF A/CORE A*], 2025

Area Chair:
- NeurIPS, ICML, ICLR, DICTA

Conference Reviewer:
- NeurIPS, ICML, ICLR, CVPR, ICCV, ECCV, KDD, ACM MM, AAAI, IJCAI, ICRA, IROS, ICPR

Journal Reviewer:
- TPAMI, Artificial Intelligence, TIP, TMM, TKDE, TCSVT, RA-L, Neural Networks, Neurocomputing, PR-L, TMLR, CIT

Guest Lecturer:
- USYD COMP 5329 Deep Learning (Week 8 - Sequential Modeling Architecture II, Week 9 - Multi-modal Foundation Model), 2026 S1
- USYD COMP 5329 Deep Learning (Week 7 - Transformer), 2025 S1
- USYD OCMP 5329 Deep Learning Online Course, 2025 S1/2025 S2/2026 S1
- USYD GenAI Summer/Winter School, 2024 S2/2025 S1/2025 S2/2026 S1

Teaching Assistant:
- Wuhan University, Introduction of Computer Networks, 2022 S1
- Wuhan University, Large-Scale Software Engineering, 2019 S1

PhD Exam Convenor:
- The University of Sydney, Chaired examination committees for 3 PhD candidates, 2026